Information Extraction from Arabic and Latin scanned invoices

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***Abstract*—The relevant entity extraction from scanned document image is a very challenging task due to highly heterogeneous templates, and several structure layouts. These problems lead to inaccuracy for document image recognized by OCR. In this paper, we propose an effective solution for these problems, in which the relevant entities are extracted from Arabic and Latin scanned invoices. The input of the system is an invoice image which is submitted to an OCR without layout analysis. After, invoices are labeled in the text recognized by the OCR. By combining the logical and physical structures, a local graph model is built for extraction entity. Finally, we implement a correction module which requires the mislabeling correction by eliminating the superfluous parts detected by labeling step. We evaluate the obtained results with 1050 real invoices as reported in experimental section.**

***Index Terms*—document image processing; information extraction; graph structure; correction module.**

I. Introduction

The automatic administrative document analysis is the most interest application in document image processing. The most task incorporated in this area contains Information Extraction (IE). The latter is defined in [1] as the process of locating and identifying an assembly of defined terms in a specific domain, ignoring other irrelevant information. We are interested in IE from OCRed Arabic and Latin invoices. The IE from multilingual sources is an obstacle which has considerably less attention than IE from mono-lingual sources. We focus on extraction entities contained in an invoice from their descriptions by a set of terms independently of their locations. To achieve this goal, certain challenges may be overcome such as non-normalized representation of the entity, Optical Character Recognition (OCR) errors, structure segmentation errors, mislabeling like superfluous tokens, and the heterogeneity of structures. The IE approaches can be categorized into three classes : context based approaches, model based approaches and structure based approaches.

The first type is based on contextual and linguistic rules for labeling elements of the entity in the text. They depend on predefined rules, domain and language. In this context, [2] has been proposed to extract events from on-line journals.

In the second type, we notice the use of Attributed Relational Graphs (ARG) [3] to represent document templates. Examples of these approaches were proposed in [4-10]. The difference between them lies in the features used for modeling the words in the model document with the same goal of creating templates. Consequently, many instances of the same document can generate very different models. Also, the regularity of ARG to correspond a document template with document instances cannot be assumed a priori.



Fig. 1. heterogeneity of structure layouts.

The structure approaches serve to benefit from the physical and logical structures representation of the document for the extraction. In this context, we have proposed, in our earlier work [11], a method for extraction entities embedded in scanned invoices. It identifies entity in unstructured data obtained by OCR without layout analysis like Tesseract engine. However,

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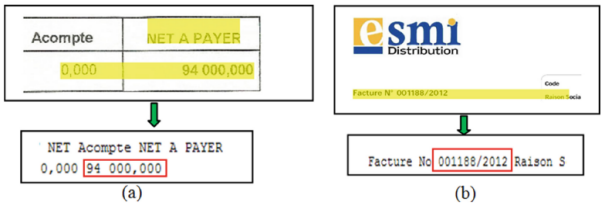


Fig. 2. Superfluous Tokens.

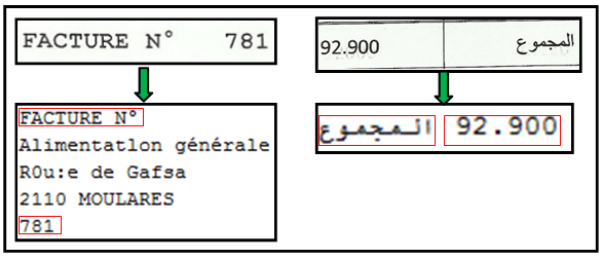


Fig. 3. Segmentation errors.

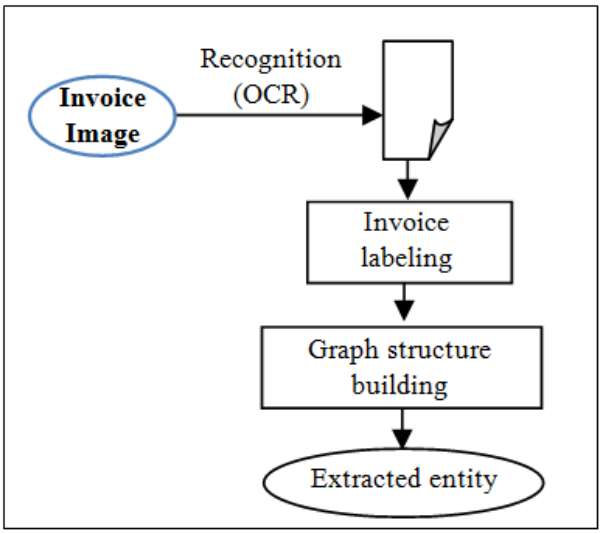


Fig. 4. Proposed System.

it does not handle the structure segmentation errors and Arabic invoices. Authors in [12,13] have proposed an automatic labeling and structure graph modeling for entity extraction and correction. However, they require high regularity of structures and automatic blocks and segments obtained by OCR.
In this paper, we propose an automatic entity labeling followed by a tokens structure graph for extraction and correction entity by elimination the superfluous tokens. This approach is strengthened by the complete absence of models matching. Also, the extraction process is robust to the heterogeneity of structure layouts as shown in Fig. 1, superfluous tokens caused by labeling errors as shown in Fig. 2, and the segmentation errors as shown in Fig. 3. The extraction task is hard, mainly when dealing with document OCRed with an OCR without layout analysis.

An overview of our proposed method and its performance is given next. In Section II, the entity extraction process is detailed. In Section III, experiments on real invoices are presented. Finally, conclusion and future work are discussed in Sections IV.

II. Entity Extraction Process

The global schema of the relevant entity extraction process is shown in Fig. 4. It is described as follows. Each incoming scanned image is OCRed. Then, entities are labeled. Ones labeled, the local structure of each entity is detected. So, a graph of relationships is built to extract and correct the relevant entity.

A. Invoice Labeling

1) Labeling: The entities are labeled using patterns of Regular Expression (Regexp), from left to right for Latin and right to left for Arabic. The only difference is the use of Unicode characters for Arabic script. For example, to label the word "fetoura" (فاتورة), we use the Unicode $\backslash u0641\backslash u0627\backslash u062A\backslash u0648\backslash u0631\backslash u0629$.
In this paper, we manipulate entities with explicit contexts and entities in association (Fig. 4 shows an example of entities in association i.e., the extraction of an entity depends on the extraction of another entity), where each invoice is defined as a set of labels $I = \{L_i\}$. A label may refer to R_i used for Reference Words (RWs) detection and F_j used for Key Fields (KFs) detection. Take the examples in Fig. 1;Facture, (N°), Total, HTVA, Date, Net, Payer, فاتورة, رقم, عدد, المجموع, التاريخ, etc are RWs. 2015/10, 201202, 153.240,000, 2014/12/17, 23/03/2015 are KFs.

2) Labels localization: T_i is the bounding box of each label on the invoice. T_i is defined as:

$$T_i \rightarrow [x_i, y_i, w_i, h_i] \quad (1)$$

With x_i, y_i represents the top-left coordinate, w_i and h_i are the width and the height respectively.

B. Graph Structure Building

1) Entity in horizontal structure: For extraction and correction an entity, we need, firstly, to define each label by a set of tokens as $L_i = SetT$. Now, $SetT$ contains two types of tokens $SetR = \{T_i^R | SetR \in SetT\}$ which is defined as the tokens of RW type, and $SetF = \{T_i^F | SetF \in SetT\}$ which is defined as the tokens of KF type.
The handling of documents recognized by an OCR suffers from several problems. Among these problems, the structure segmentation errors i.e., the physical structure of the entity in the image does not correspond to its logical structure in the text as shown in the example in Fig. 3. Then, we obtain two cases of entity in horizontal structure.
In the first case, we create an input syntactic pattern to identify the SetR followed by SetF.
In the second case, we need to establish a geometric relationships between the R_i and the F_j to extract the relevant entity. This geometric relation is inspired by the notion of visibility [14]. This notion makes it possible to link two objects such that another object does not intersect this link. In our case,

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Fig. 5. Entities in association.



Fig. 6. Geometric relationship of a horizontal structure.

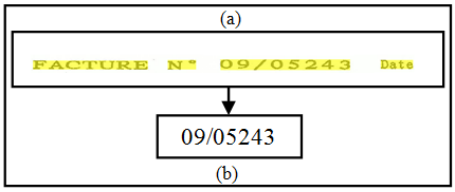


Fig. 7. (a) Entity with superfluous token. (b) Relevant token.

we consider that a RW is related to the relevant KF if there is some horizontal visibility between the two. We add a visibility threshold to check if the RW and KF are sufficiently visible. In Fig. 7, we present an example of an established geometric relation to extract a horizontal entity.

2) Correction Entity in horizontal structure: We propose a correction module for elimination superfluous tokens. It represents the arrangement of relevant tokens of local entity. For every token T_i^F, we find the right nearest neighbor. Then, based on this neighbor, we measure the distance d_{ij} between them. $T_i^F(j = i + 1)$ is called nearest neighbor if $d_{ij} \leq S$. S is the threshold defined as the maximum distance between two consecutive tokens. This threshold is empirically defined. In Fig. 5(a), the "FACTURE" and "N°" are deleted. The KFs contain the superfluous token "Date". After applying the correction module, only the relevant token "09/05243" is extracted.

3) Graph building for vertical structure:

Labels filtering: For extraction entity in vertical structure, we need to establish a structural relationships between R_i and F_j. Since the drawing of relationships between R_i and all the F_j is time consuming and no avail, we propose to filter the labels. We keep only the candidate labels inside an Area Of Interest (AOI) created under the bounding box of R_i. The area AOI is created empirically in which the relevant entity does not exceed the limit of this AOI.

Graph building for relevant entity extraction: A graph is built $G = (N, O, Q)$ in which N is a node of the label R_i. O is a set of finite nodes that represent the labels F_j having the bounding box inside the AOI. $Q \subseteq N \times O$ is a finite set of arcs which represent a geometric relationships between the node N and O. Each arc $q_{ij} \in Q$ relating the node N and o_j is represented by No_j. We define a feature vector which describes the geometric relationships between N and o_j inside the AOI.

$$a_{ij} = (CN_i, Co_j, q_{ij}) \tag{2}$$

Where: CN_i is the center of the node N. Co_j is the center of each node o_j. q_{ij} is the whitespace (distance) that separates

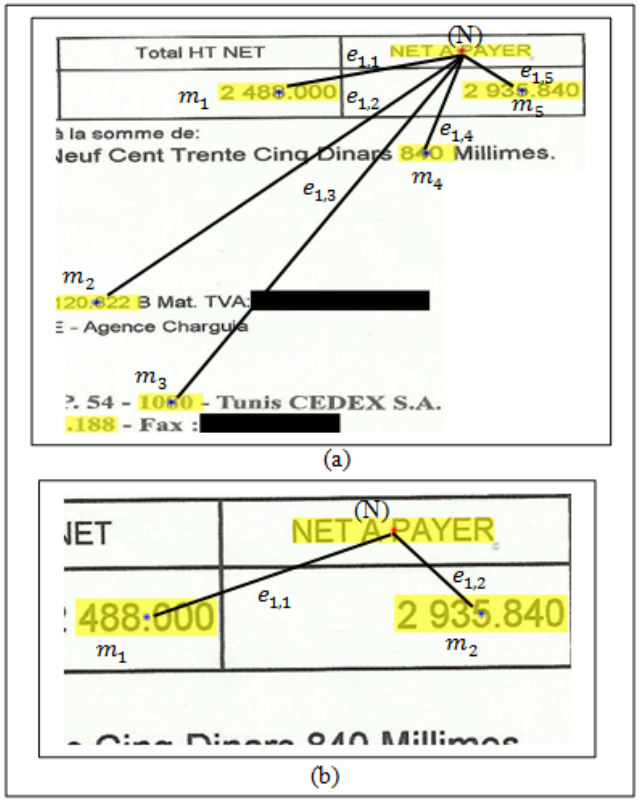


Fig. 8. Determination of labels inside AOI. (a) Before AOI. (b) After AOI.

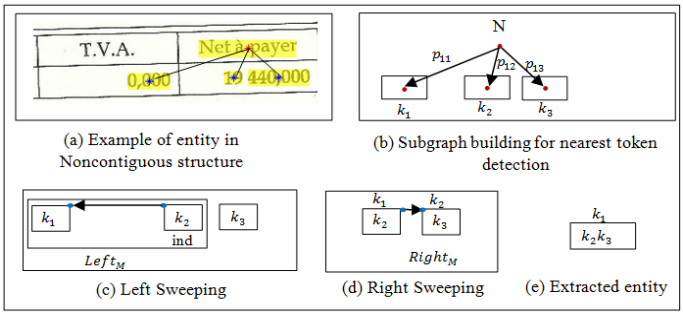
the the labels corresponding to N and o_j, as we can view in Fig. 5. The idea is to detect the nearest o_j to N (steps 8-11 algorithm 2). We consider only the nodes having the centers inside the AOI. In Fig. 6, m_2 represents the nearest o_j to N i.e., m_2 is the relevant entity. However, the latter may contain noisy tokens that must be eliminated. To achieve this goal, a correction module is proposed.

Subgraph building for entity correction : The correction of an entity in verticals is more complex than that in horizontal structure. We need to find the right nearest neighbors and the left nearest neighbors.

- Nearest of the nearest determination: In this step, a subgraph of relationships, as described in algorithm 2, is built between the node N and the tokens $K = \{k_v\}$ of the nearest node o_j. The nearest token is the frame used for a sweeping to the right and then to the left to determinate the relevant tokens. To determinate the nearest token, we calculate the distance p_{iv} between the node N and each token k_v. The nearest token possesses the minimum distance with N. We call this token "ind" as it represents an index from which begins the sweeping. In Fig. 7, the "ind" is k_2.
- Sweeping: The sweeping is the exploration token by token of an entity. It is done in both directions to the left and then to the right for superfluous tokens elimination.

The left sweeping consists in creating a matrix containing a set of finite tokens. The geometrical relationships, provided by the distances measured between tokens inside $Left_M$, are used to

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(a) Example of entity in Noncontiguous structure
(b) Subgraph building for nearest token detection
(c) Left Sweeping
(d) Right Sweeping
(e) Extracted entity

Fig. 9. Subgraph of tokens.

concatenate relevant tokens. This matrix is defined as :

$$Left_M = (k(1 : ind)) \tag{3}$$

Whenever, we calculate a distances between each two consecutive tokens. This distance must not exceed the threshold S previously identified (explained in Section II.B.2). The distance is calculated as:

$$n_{z,z-1} = k_{(z,1)} - (k_{(z-1,1)} + k_{(z-1,3)}) \tag{4}$$

The left sweeping outcome is $Left_M$ containing only one element grouping the relevant tokens. This element is added to the beginning of the created matrix $Right_M$ for the right sweeping. So, all relevant tokens are grouped in $Right_M$ which is defined as:

$$Right_M = (K(Left_M + 1 : end)) \tag{5}$$

The concatenation in the right sweeping is done with the same principles detailed in the left sweeping. Fig. 6 shows the process of sweeping for tokens concatenation. In Fig. 7(b), a subgraph of geometric relationships is established between the nodes. The nearest token, ind, having the minimum distance with the node N is detected. The latter is "19". In Fig. 7(c), $Left_M$ contains two tokens "0,000" and "19". The distance between these tokens exceeds the threshold S. For that, the token "0,000" is eliminated. In Fig. 7(d), $Left_M$, containing one element, is integrated in the start of $Right_M$ and the right sweeping begins. In Fig. 7(e), the right sweeping allows the concatenating of relevant tokens ("19", "440,000").

III. EXPERIMENTAL RESULTS

A. Dataset

Our method is experimented on a corpus containing 1050, multi-font, multi-size, Arabic and Latin real invoice images. These invoices are scanned from different companies.The entities extracted from invoices are of type invoice number (N°), invoice date (DT), Pre-tax Amount (PA), Total Including Tax (IT), Holdback (H), Account Identity (AI) and Balance Due (BA). It is important to indicate that our system sustains the data extraction from grayscale, color and bi-tonal (black and white) images. Also, The manipulated invoices contain graphical elements, logos, vertical and horizontal lines, and tables. Some manipulated invoices are shown in Fig. 8. For evaluation, we use ground truth tables manually prepared.

Algorithm 2: Extraction and correction entity in vertical structure

Input: N // RWs
O // KFs
Output: RT// Relevant tokens
begin
  CN_i = Calculate_Center(N_i)
  AOI = $Region_under_N$
  for each o_j in AOI do
    Co_i = Calculate_Center(o_i)
    $q_{ij} = Euclediandist(CN_i, Co_j)$
  end
$RE = o_j(dmin(q_{ij}))$ // The relevant entity is the o_j having the minimum distance with N.
$RE = K$ // Tokens of the relevant entity
  for each k_v do
  $a_{iv} = Euclediandist(CN_i, Ck_v)$
  end
  $ind = k_v(dmin(a_{iv}))$
$lm = size(Left_M)$
  for $j = lm : 1 : -1$ do
    if $n_{z,z-1} \leq S$ then
      $k_z = concat(k_z, k_{z-1})$
      $k_{z-1} = []$
    elseif $n_{z,z-1} > S$ then
      $Left_M = k_z$
    end
  end
$Right_M = (K(Left_M + 1 : end))$
$rm = size(Right_M)$
  for $q = 1 : rm$ do
    if $n_{z,z+1} \leq S$ then
      $k_z = concat(k_z, k_{z+1})$
      $k_{z+1} = []$
    elseif $n_{z,z+1} > S$ then
      $Right_M = K$
    end
forend
$RT = K$
end

B. Evaluation

Geometric relationships are very efficient for entity extraction, regardless to the mismatch between unstructured data obtained by the OCR and its physical representation in the invoice image. Our system is able to detect three types of structures detailed in TABLE I (V: Vertical, H: Horizontal). Fro example, H/H means the ratio physical structure / Logical structure. The first and second type are explained in our earlier work [11]. The third type, which is the main of this work, is caused by segmentation errors. Our system has shown its robustness in extracting entities even with this type of errors.

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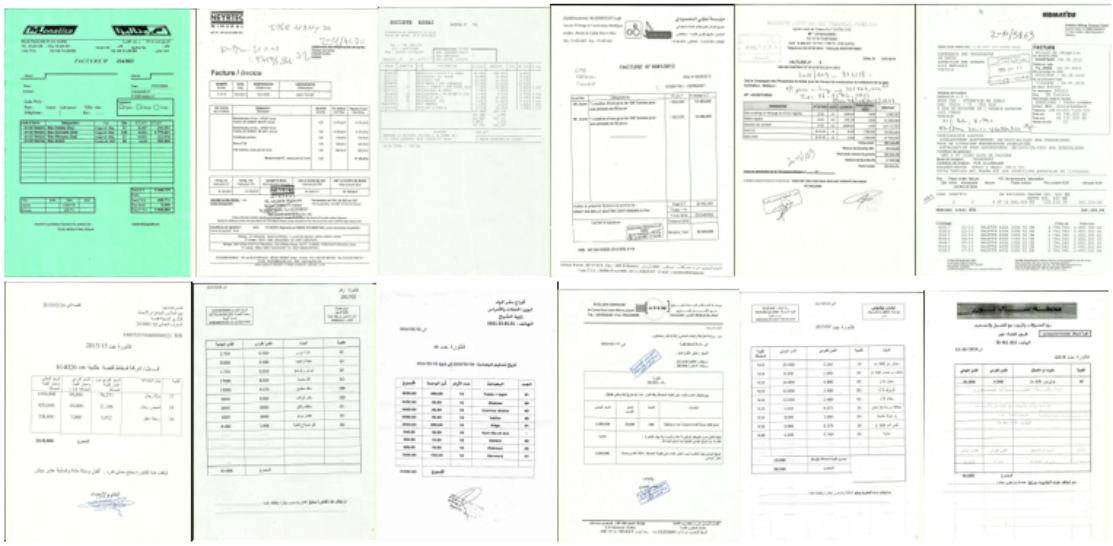


Fig. 10. Sample of real invoices in our dataset.

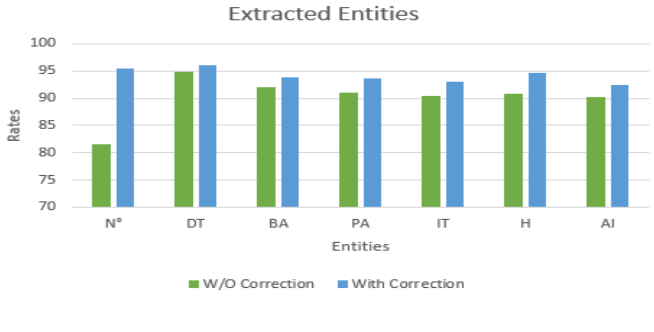


Fig. 11. Extraction results of the entities.

TABLE I
EXTRACTED STRUCTURES

Entity type	Structure		
	H/H	V/V	H/V
N°	936	55	9
Date	898	97	5
BA	921	63	16
PA	916	61	23
IT	921	62	17
AI	817	15	-
H	66	-	-

TABLE II
IMPACT OF FILTERING

	Run time (sec/inv)
Without filtering	0.849
With filtering	0.725

The labels filtering step has greatly reduced the candidate entities. We can see in TABLE II a decrease in the time consuming due to this filtering.

The correction module is integrated for superfluous tokens elimination. The goal is to increase the accuracy of the extraction process. To show the impact of this module, we use two options; without correction (W/o C) and with correction (With C), as shown in Fig. 11. It demonstrates the extraction results of our local structure analysis for information extraction and correction. We achieve rates up of 92%. The most interesting result is for the N° entity since it has a countless number of formats. The obtained rates show the power possessed by our method of extraction and correction.

Despite its efficiency, the system lost many results due to the OCR errors especially in Arabic invoices because of their complex characteristics. As it is known, the success of extraction process depends heavily on the accuracy of text recognition system. Since Arabic text is cursive, it is not easy to recognize it.

IV. CONCLUSION AND FUTURE WORK

An entity extraction approach in scanned invoices recognized by OCR, is described in this paper. Our system has shown its robustness even for structure segmentation errors. The structural relationships and the graphs built for local entity labels have proved to be efficient in improving the extraction process. The correction module based on superfluous tokens elimination was effective in increasing the precision of the extraction process. However, our system suffers from some erroneous results. This failure is due in large portion to OCR errors especially in Arabic invoices. For that, we propose, in future work, to integrate a preprocessing step which improves the accuracy rate within the OCR process and enhances the text recognition process and subsequently the extraction process.

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REFERENCES

[1] R. Hashemi, C. W. Ford, , T. Vamprooyan and J. Talburt, *Extraction of features with unstructured representation from html documents*. IADIS International Conference on WWW/Internet, pp. 47-53, 2002.

[2] J. Piskorski and R. Yangarber, *Information extraction: Past, present and future*. Multi-source, Multilingual Information Extraction and Summarization, pp 23-49, 2013.

[3] M. A. Eshera and K. S. Fu, *An image understanding system using attributed symbolic representation and inexact graph-matching*. IEEE Trans Pattern Anal Mach Intell, pages 604-618, 1986.

[4] F. Cesarini, M. Gori, S. Marinai and G. Soda, *INFORMys: a flexible invoice-like from-reader system*. IEEE Trans Pattern Anal Mach Intell, pages 730-745, 1998.

[5] Y. Yshitani, *Model-based information extraction method tolerant of OCR errors for document images*. International Conference on Document Analysis and Recognition, 2001.

[6] Y. Yshitani, *Model-based information extraction and its applications for documents images*. Workshop on document layout interpretation and its applications, 2001.

[7] P. Duygulu and V. Ataly, *A hierarchical representation of form documents for identification and retrieval*. Int J Doc Anal Recognit, pages 17-27, 2002.

[8] F. Schulz, M. Ebbecke, M. Gillmann, B. Adrian, S. Agne and A. Dengel *Seizing the Treasure: Transferring Layout Knowledge in Invoice Analysis*. Applied Intelligence, 2012.

[9] A. Belaid, VD Poulain, H. Hamza and Y. Belaid, *Administrative document analysis and structure*. Biba M(ed) Learning structure and schemas from documents. Springer, Berlin, 2011.

[10] C. Peanho, H. Stagni and F. da. Silva, *Semantic information extraction from images of complex documents*. Applied Intelligence, 2012.

[11] N. Rahal, M. Benjlaiel and M. A. Alimi, *Entity Extraction and Correction based on Token Structure Model Generation*. S+SSPR, 2016.

[12] H. Dejean, *Extracting Structured Data from Unstructured Document with incomplete Resources*. International Conference on Document Analysis and Recognition, 2015.

[13] N. Kooli and A. Belaid, *Entity Matching in OCRed Documents with Structured Databases*. International Conference on Pattern Recognition Applications and Methods, pp. 165-172, 2015.

[14] S. K. Wismath, *Computing the full visibility graph of a set of line segments*. Information processing letters, pp. 257-261, 1992.

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